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Mobile Architect & Programming

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Project Two: Read Me

Overview:

This purpose of this project is to create a dashboard that allows the client Grazioso Salvare to interact with the Animal Center Outcomes database. The dashboard allows any user to filter animals based on established rescue categories andsee the filtered results through a dynamic pie chart and map.

Functionality:

Interactive Filtering:

Users can filter the dataset using a dropdown menu that shows the following rescue types:

- Water Rescue
- Mountain/Wilderness Rescue
- Disaster Rescue or Individual Tracking

There is also a Reset button below the dropdown menu to reset row selection. Rescue filtering was created by coding a conditional statement that would filter animals based on animal type, calculated age, and breed.

Data Table

The dashboard contains a data table displaying records from the dataset. It supports pagination, with 10 rows so as not to overwhelm the user. Uuser's can filter individual columns, and display map results with the single row selection. The table updates dynamically when a filter option is selected.

Map

There is a geolocation map provided below the data table on the right-hand side. It displays the locations of the animals using a larger pointed marker so user's can quickly identify the location of animals.

Features:

- Dynamic: Markers are displayed when the user selects one of 3 filters: Water Rescue, Mountain/Wilderness Rescue, or Disaster Rescue/Individual Tracking. To reduce lag, the map does not show any location markers on load or on the "All" filter.
- Single Row: Selecting a single row on the table shows the animal's respective location on the map, and no other marker.
- Reset: The reset button resets the filter, and by extension, the map.

Pie Chart

The pie chart is located beneath the data table, on the left hand side. It shows the distribution of dog breeds in the dataset.

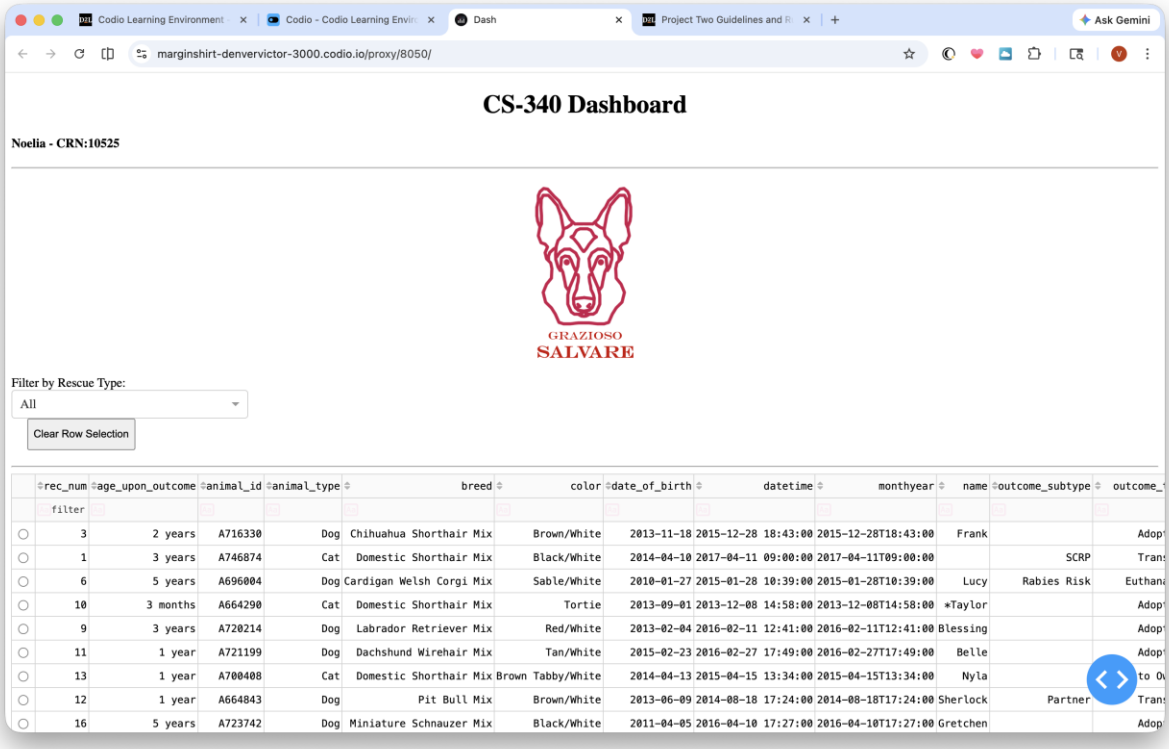
Features:

- Simplification: The pie chart shows only the top 15 breeds on load or on the “All” filter. This was because there over 30 breeds of animals, and the pie chart became deformed and oblong. It groups less common breeds under “Other”.
- Dynamic: It automatically updates when the filters change or when the reset button is hit to reflect the data in the data table.

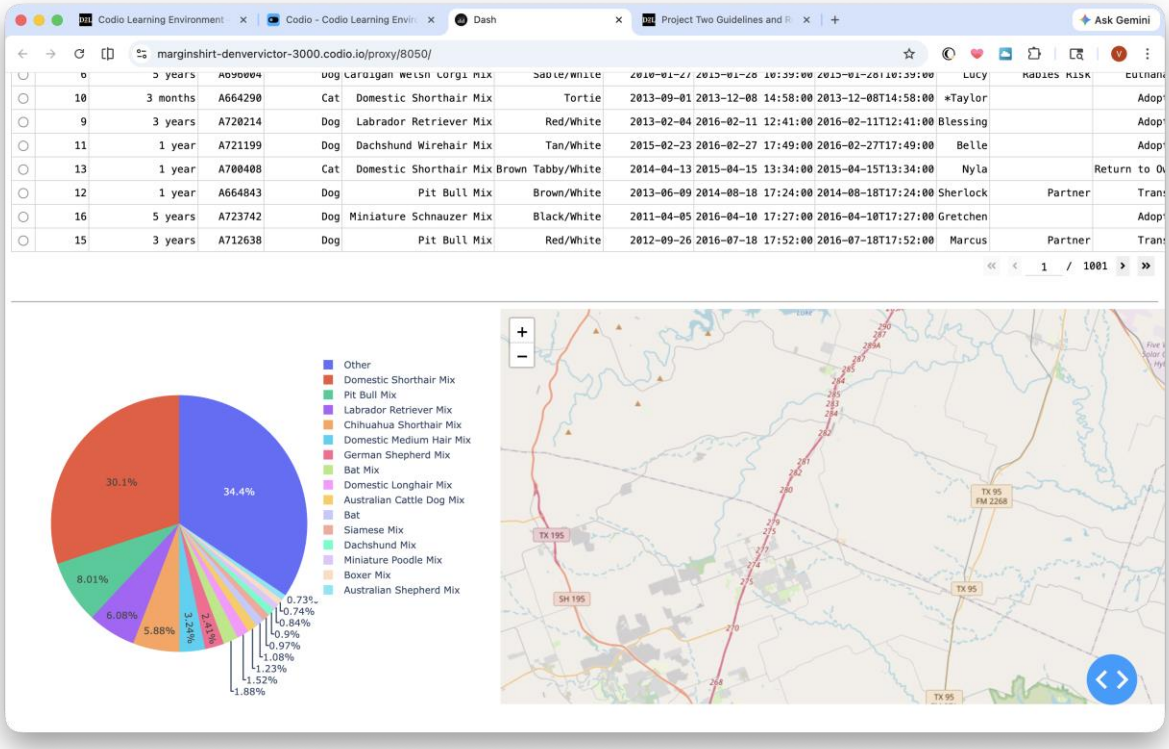
Dashboard Views

The following are screenshots provided to give you a clearer picture of the dashboard.

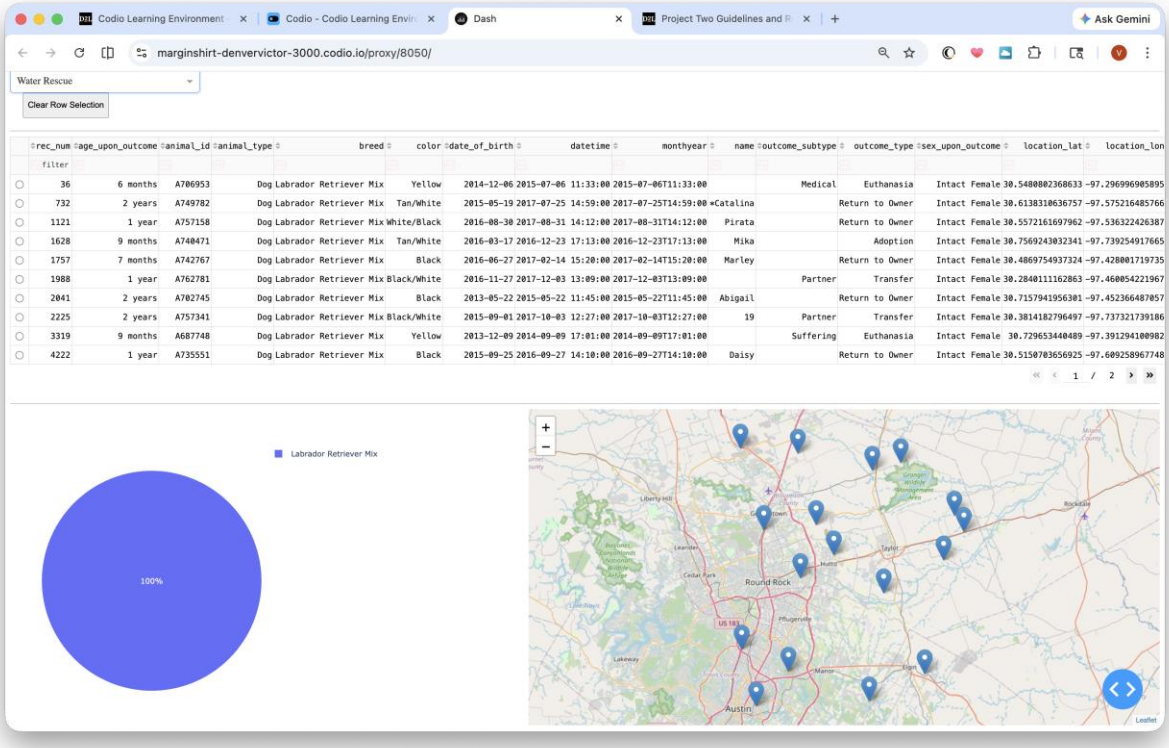
On Load



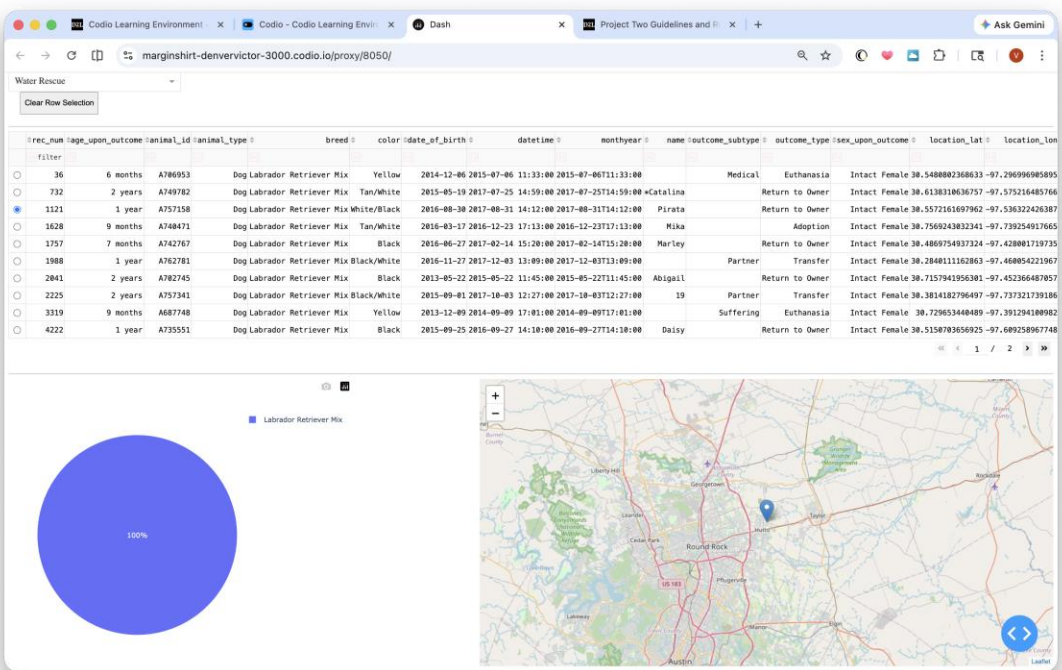
rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_
filter											
3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adop
1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Tran
6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	Rabies Risk	Euthan
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adop
9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adop
11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adop
13	1 year	A700408	Cat	Domestic Shorthair Mix	Brown Tabby/White	2014-04-13	2015-04-15 13:34:00	2015-04-15T13:34:00	Nyla		to O
12	1 year	A664843	Dog	Pit Bull Mix	Brown/White	2013-06-09	2014-08-18 17:24:00	2014-08-18T17:24:00	Sherlock	Partner	Tran
16	5 years	A723742	Dog	Miniature Schnauzer Mix	Black/White	2011-04-05	2016-04-10 17:27:00	2016-04-10T17:27:00	Gretchen		Adop



Water Rescue Filter



rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_lon
filter														
36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female	30.5480802368633	-97.296996905895
732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intact Female	30.6138310636757	-97.575216485766
1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intact Female	30.5572161697962	-97.536322426387
1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intact Female	30.7569243032341	-97.739254917665
1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intact Female	30.4869754937324	-97.428001719735
1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intact Female	30.2840111162863	-97.460054221967
2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intact Female	30.7157941956301	-97.452366487057
2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12:27:00	2017-10-03T12:27:00	19	Partner	Transfer	Intact Female	30.3814182796497	-97.737321739186
3319	9 months	A687748	Dog	Labrador Retriever Mix	Yellow	2013-12-09	2014-09-09 17:01:00	2014-09-09T17:01:00		Suffering	Euthanasia	Intact Female	30.729653440489	-97.391294100982
4222	1 year	A735551	Dog	Labrador Retriever Mix	Black	2015-09-25	2016-09-27 14:10:00	2016-09-27T14:10:00	Daisy		Return to Owner	Intact Female	30.5150703656925	-97.609258967748



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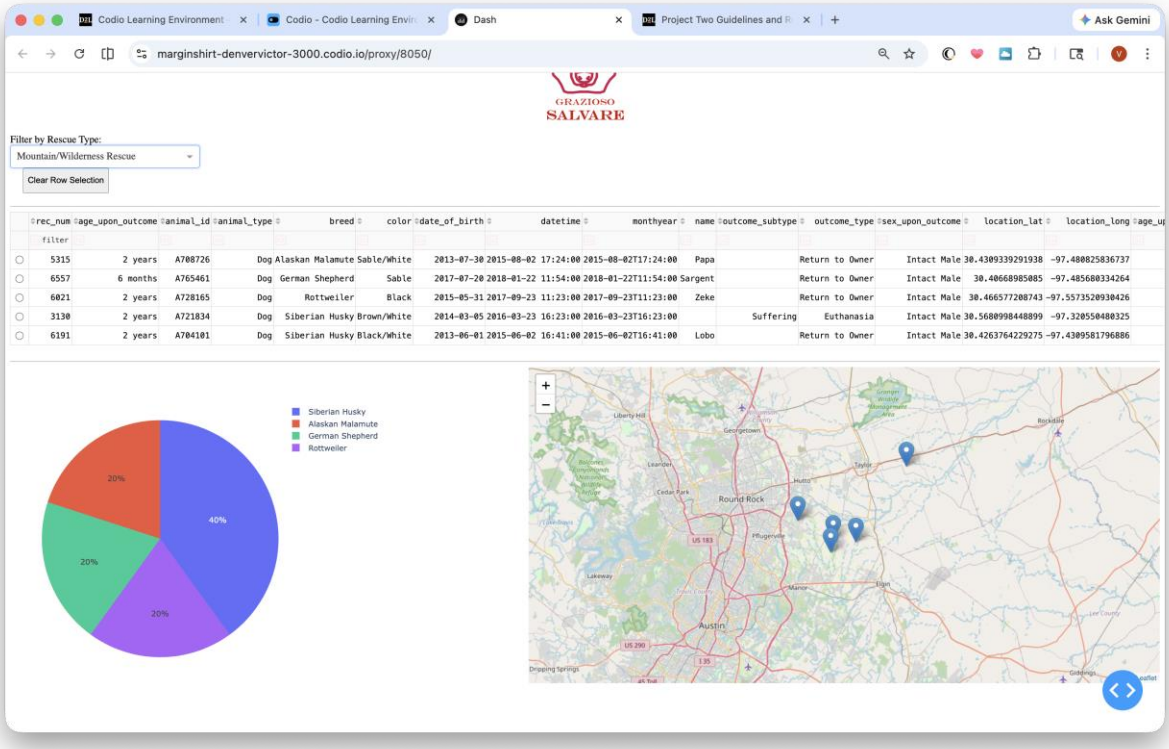
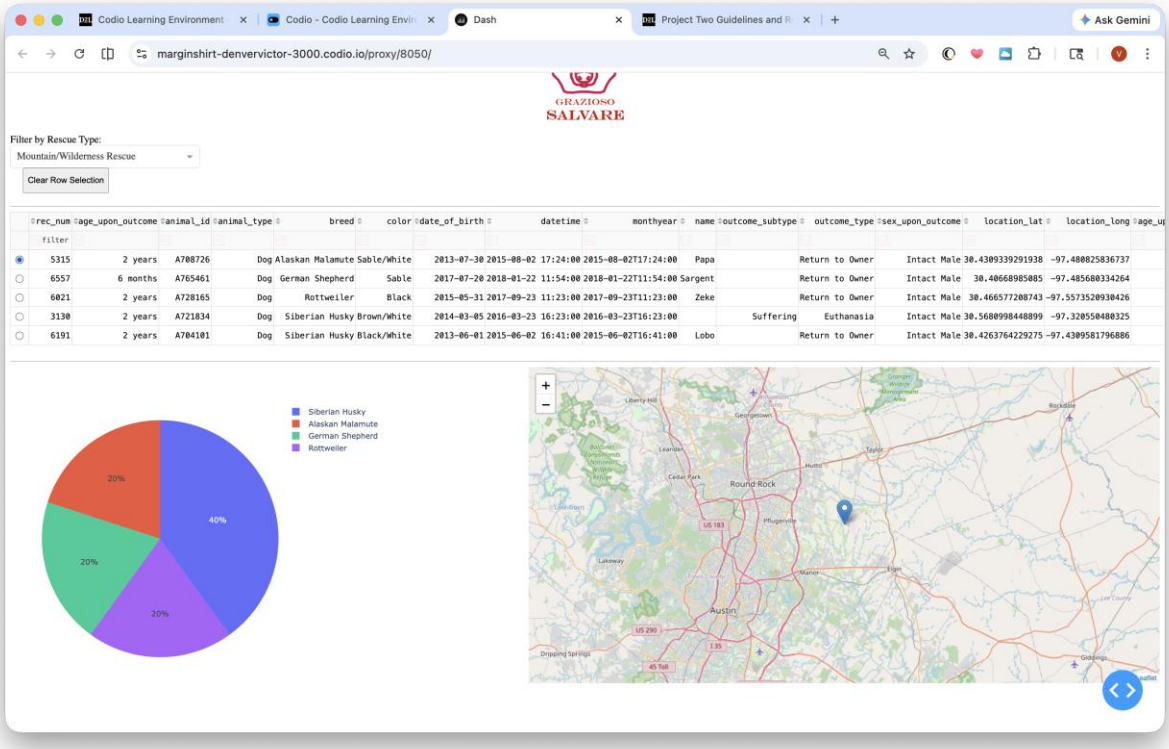
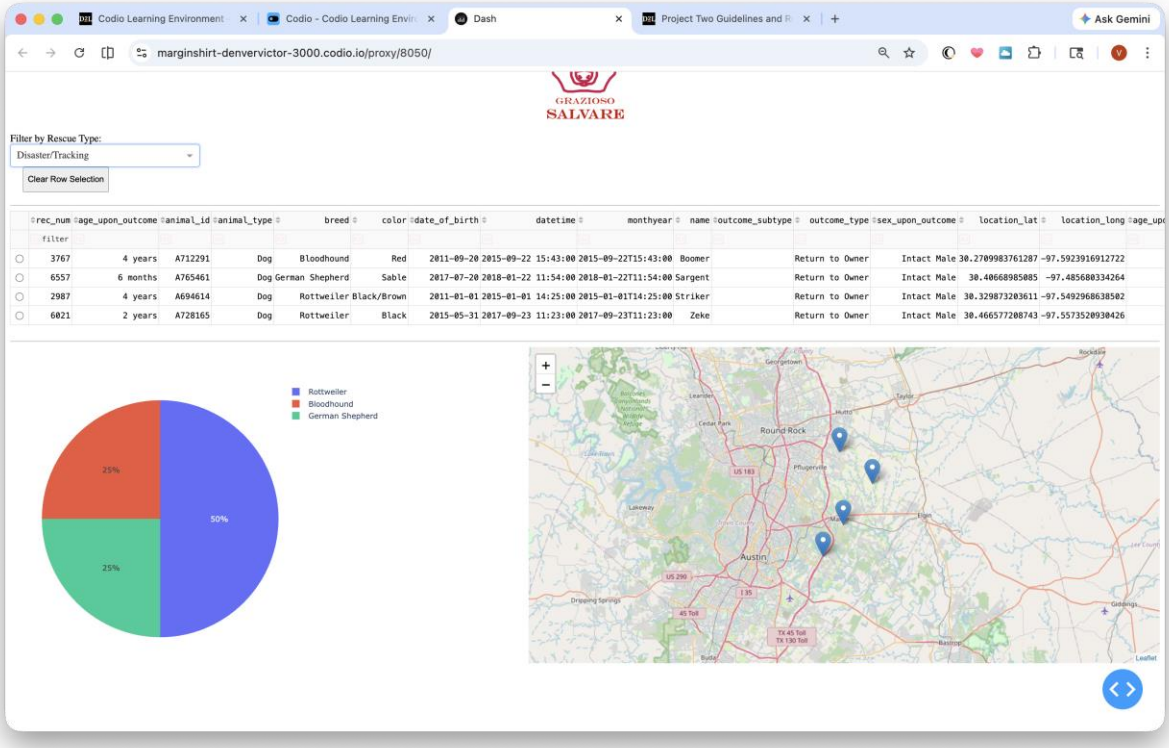
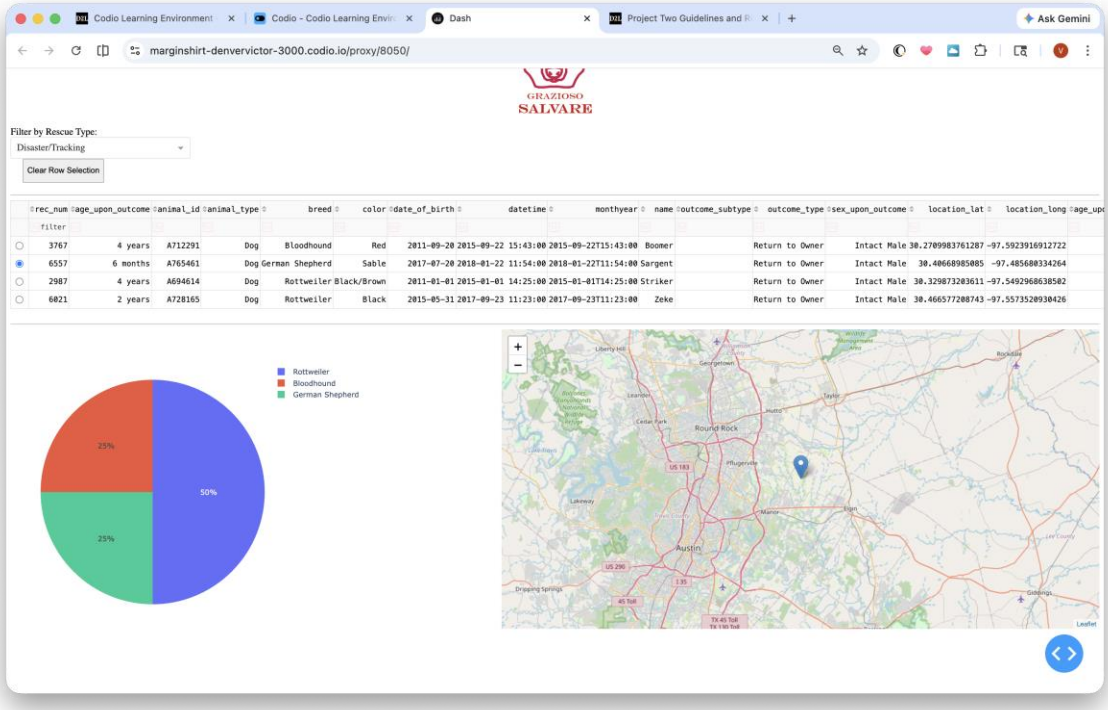
Mountain/Wilderness Rescue Filter



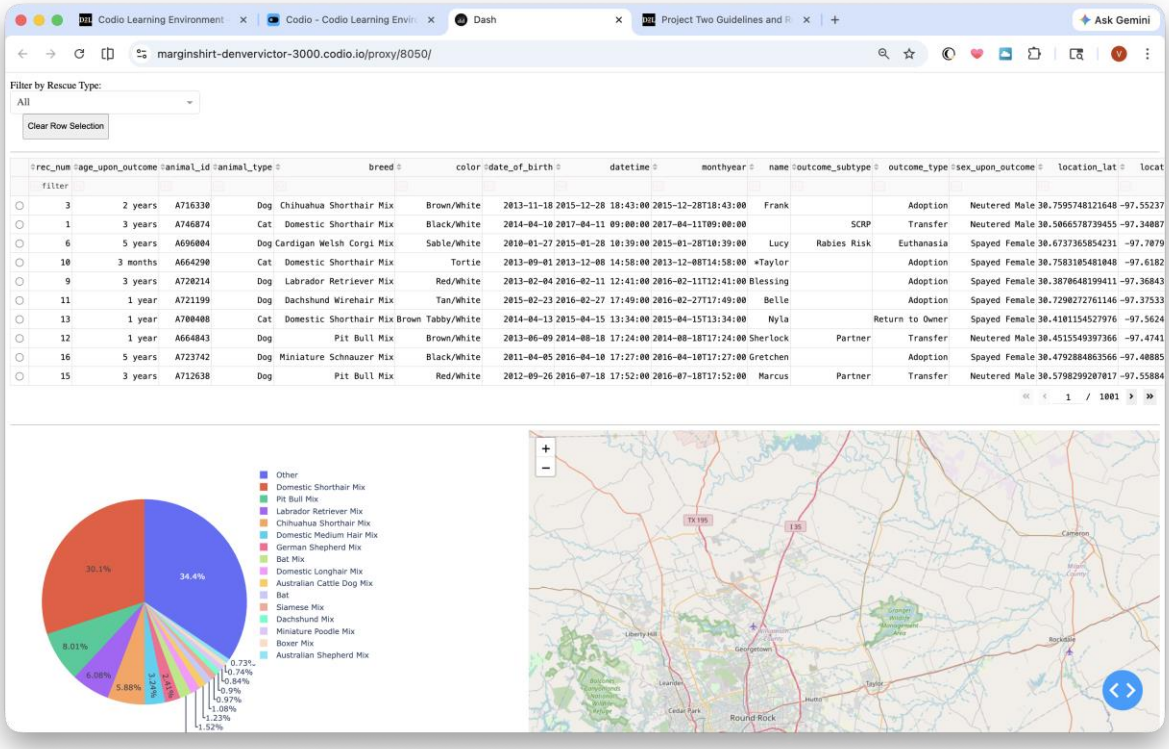


Disaster Rescue and Individual Tracking Filter





Reset + All



Tools and Technologies Used

I used the following tools to create the dashboard.

- Python: Primary programming language used for both the Project Two dashboard and for the CRUD module. Python connects the dashboard components, processes the data, and communicates with the MongoDB database.
    - Website: https://www.mongodb.com/
- MongoDB: This was the database and model component. It stores the animal shelter data for the dashboard to retrieve and display. Because of the way MongoDB stores its information (JSON-like), Python can easily access it through libraries like PyMongo. Thanks to this, the CRUD module can create, read, update, and delete data records in the MongoDB database.
    - Website: https://www.mongodb.com/
- Dash: This framework provides the view and controller parts of the dhasboard by allowing me to create a user interface that the user can interact with.
    - Website: https://dash.plotly.com/
- Plotly: Plotly was used to generate the pie chart visuals on the dashboard. Since it integrates with Dash easily, I was able to make the graph dynamic.
    - Website: https://plotly.com/python/

- Dash Leaflet: This was used to display the map element of the dashboard. It takes location data from the dataset and plots it on an interactive map within Dash.
    - Website: https://dash-leaflet.herokuapp.com/
- Pandas: This library was used for data manipulation and formatting. It converts the database into something that could be filtered easily and displayed on the dashboard.
    - Website: https://pandas.pydata.org/
- Numpy: This library was used to support numerical operations and data handling in Python.
    - Website: https://numpy.org/

Steps

Step One: Database

The Austin Animal Center Outcomes dataset was loaded into MongoDB within the Codio environment. A CRUD Python module was created to allow Python applications to interact with the MongoDB database.

Step Two: Dashboard

A dashboard template provided in the course materials was opened in the Jupyter Notebook environment in Codio. The template already included the basic structure for the Dash application.

Step 3: Code Correction

The template code was modified and debugged so that it could correctly retrieve data from the MongoDB database using the CRUD module. This included updating the CRUD import.

Step 4: Visuals

A pie chart and map were added. The pie chart shows breed distribution across several filters.

Step 5: Interactive Filter

A dropdwon filter was added to let users select between the different rescue categories. Dash callbacks dynamically update the data table and subsequent elements (pie chart and map) based on the filter chosen.

Step 7: Testing

The dhasboard code was executed and tested. Several bugs appeared but I went back and debugged them over the course of 2hrs.

Challenges:

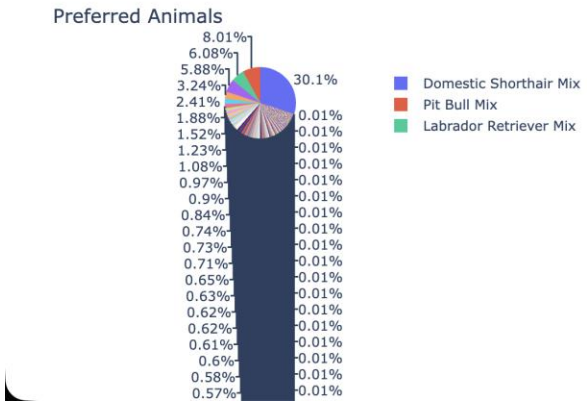
Dash Callback

This was an error I encountered frequently. Dash doesn't allow multiple callbacks so I had to combine two separate functions into one.

MongoDB ObjectID Compatibility

MongoDB automatically includes an _id field with an ObjectID type. This type caused compatibility issues with the Dash DataTable. So, I removed the _id field before displaying the data table.

Pie Chart:

My pie chart was "deformed" due to the sheer number of breeds, because the dataset includes more than cats and dogs. The lines to point to a particular slice of the pie chart had overlapped into one shape. My solution was to implement a limit of 15 breeds.

Screenshot of deformity: